

Roshan Dharan

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Technical Skills

AI / Agent Systems: LangGraph, RAG (LangChain, ChromaDB), LLM fine-tuning (LoRA), prompt engineering, eval/QA pipelines, XGBoost, SHAP, PyTorch, Transformers
Languages: Python, Java, C++, JavaScript, TypeScript, SQL, HTML/CSS, MATLAB
Backend, APIs & Systems: FastAPI, Spring Boot, Node.js, REST API design, distributed systems, unit testing (Mockito), root-cause debugging
Cloud & Databases: AWS (S3, EC2, Lambda), GCP, OCI, Snowflake, PostgreSQL, MongoDB, Redis
DevOps & Tools: Git, Docker, Kubernetes, CI/CD (GitHub Actions), Agile/Scrum, Linux, JIRA

Experience

Keck Medicine of USC | Software Engineer Intern Los Angeles, CA | Nov 2025 - Present

- Led a 6-person team as Technical Lead, owning the full SDLC end-to-end - requirements, design, coding, testing, and deployment - for a production predictive system spanning Snowflake, two versioned XGBoost models, and a REST API integration.
- Shipped real-time no-show risk scores across tens of thousands of monthly transactions, directly informing scheduling decisions in Cerner RevCycle, a mission-critical clinical system.
- Fine-tuned a transformer-based NLP model (LoRA on a T5/BART backbone) and built an automated evaluation/QA pipeline that caught defects across 2,000+ generated outputs pre-deployment, cutting manual review effort by approximately 30%.
- Piloted a real-time capacity-tracking service in Keck's PACU, outperforming the existing tool and projecting \$1M+ in annualized savings (approximately \$1,000/bed/day); presented to hospital leadership and at a Spring 2026 internal hackathon.

Oracle | Associate Software Engineer Bengaluru, India | Oct 2023 - Jul 2025

- Architected scalable, concurrent backend pipelines processing 600,000 records daily, troubleshooting issues across server-side and client-side layers to deliver reporting infrastructure that informed \$200M in decisions across 17 departments.
- Developed an automated data-extract service with third-party system integration, applying efficient algorithmic design to cut operating costs by \$12M while reclaiming 114 hours of manual effort annually.
- Built a real-time monitoring backend integrating multiple external APIs, automating 500 weekly manual checks and eliminating 15 hours of manual work per week.
- Optimized cloud-based reporting infrastructure across 17 departments, engineering backend ETL workflows that cut report generation time by 40% and improved system reliability.

Oracle | Software Engineering Intern Bengaluru, India | Jan 2023 - Jul 2023

- Modernized five on-premise BI systems to Snowflake and built an 8-thread Python/Selenium engine, reducing dashboard load time by 35% and manual effort from 44 hours to under 3 minutes.

AI Agent Projects

Real-Time AI Command Center LangGraph, XGBoost, SHAP, ChromaDB, FastAPI | Mar - Jul 2026

- Deployed a real-time multi-agent system (LangGraph StateGraph, conditional routing) firing on every incoming event and running five XGBoost classifiers with SHAP explainability at 0.77-0.83 validation AUC.
- Built an LLM-powered decision agent that synthesizes five model outputs and a ChromaDB retrieval layer over 10 reference documents into structured JSON action plans, logged to PostgreSQL and broadcast live via SSE with Accept/Override/Escalate attribution.

RAG-Powered Codebase Q&A Agent LangChain, ChromaDB, sentence-transformers | Feb - Apr 2026

- Built a retrieval-augmented generation agent using language-aware chunking and MMR retrieval over locally embedded vector search, ingesting Flask's 88 files into 573 chunks in 5.8 seconds to answer cited codebase questions.
- Ran a 10-query retrieval evaluation, tuning MMR to raise hit-rate@k from 70% to 90%, then hardened the system with git-diff incremental re-ingestion, a SQLite-backed repo registry, and size/timeout guards.

Education

University of Southern California Aug 2025 - May 2027

M.S. in Computer Science GPA: 3.67/4 | Los Angeles, CA

Coursework: CSCI 570, CSCI 572, DSCI 552, CSCI 576

REVA University Jul 2019 - Jul 2023

B.Tech. in Computer Science and Engineering GPA: 3.97/4 | Bengaluru, India

Coursework: Data Structures & Algorithms, OOP, Advanced DBMS, Operating Systems, Machine Learning, Artificial Intelligence, Cloud Computing, Big Data, Data Science (R)

Honors & Awards

Oracle Pinnacle Award - 145.83% TAG, 292% productivity; solution standardized org-wide. Jun 2025

Oracle Champions of Service Award - Recognized for outstanding customer service. Mar 2025